Practice Problems for Chapters 1&2

Chapter 1

1. Find the natural domain and range of each of the following functions.
   (a) $f(x) = \dfrac{x-2}{x^2-4}$
   (b) $f(\theta) = \csc\theta$
   (c) $f(x) = \sqrt{x - 3x^2}$

2. Find the natural domain of $f(x) = \dfrac{\sqrt{x-2}}{\sqrt[3]{x}-2}$.

3. Let $f(x) = |x| - 3x + 3$. Express $f(x)$ in piecewise form without using absolute values and graph it.

4. Let $f(x) = \sqrt{x^2-9}$ and $g(x) = \sqrt{x+4}$, find
   (a) the natural domain of $\dfrac{g}{f}$.
   (b) $f \circ g(x)$.

5. Sketch the graph of following functions.
   (a)$f(x) = \dfrac{1}{x-2}$
   (b)$f(x) = \sqrt{x+4}$.
   (c)$f(x) = 2\sin(2x - \pi)$.

6. Find $\sin^{-1}\left(\sin\dfrac{6\pi}{7}\right)$.

7. Simplify $2^5 \cdot 2^{-3} + 4^{-0.5} + 8^{2/3} + \log_2 8$.

8. Solve for x in:
   (a) $\ln 4x - 3\ln(x^2) = \ln 2$ for x.
   (b) $\ln(1/x) = -2$.
   (c) $e^x - 3xe^x = 0$.
   (d) $e^{-2x} - 3e^{-x} = -2$.

9. Consider the parametric curve $x = \sqrt{t} - 2$ and $y = 3t + 4$. Express this curve in the form of $y = f(x)$.

10. Find paramteric equations for the ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{9} = 1$.

Chapter 2

1. Find the following limits:
   (a)$\lim_{x\to 6} \dfrac{x+6}{x^2+36}$.
   (b)$\lim_{x\to 2^+} \dfrac{x-7}{x-2}$.

(c)$\lim_{x\to -3} \dfrac{x^2-9}{x^2+x-6}$.

(d)$\lim_{x\to -\infty} \dfrac{x^2-9}{x^2+x-6}$.

(e) $\lim_{x\to 1^-} \dfrac{(x+1)(x+2)}{x-1}$.

(f) $\lim_{x\to \frac{\pi}{2}^+} \tan x$.

(g) $\lim_{x\to -\infty}(\dfrac{3}{x^2}+e^x)$.

(h)$\lim_{x\to +\infty} \sqrt[3]{\dfrac{3x^7-4x^2}{2x^7+1}}$.

2. Find $\lim_{x\to +\infty} \sqrt{x^2+3x}-x$.

3. If the line $x=3$ is a vertical asympotote of the graph of $y=f(x)$, what can you say?

4. Find the horizontal asymptotes, if any for $y=\dfrac{x^3-x^2+10}{3x^2-4x}$.

5. Using the rigorous definiton to prove that $\lim_{x\to 1}(7x-5)=2$.

6. Find values of x, if any, at which the function $f(x)$ is not continuous.

   (a) $f(x)=\dfrac{5}{\sqrt{x}}+\dfrac{2x}{x-4}$.

   (b) $f(x)=4-\dfrac{8}{x^4+x}$.

   (c) $f(x)=\sin(\dfrac{x}{x-5})$.

   (d) $f(x)=\dfrac{1}{1-2\sin x}$.

7. Find values of x, if any, at which the function $f(x)$ is not continuous, and determine whether each such value is a removable discontinuity.

   (a)$f(x)=\dfrac{|x|}{x}$.

   (b)$f(x)=\dfrac{x-1}{x^2-1}$.

8. Find the limits.

   (a) $\lim_{x\to +\infty} \cos(\dfrac{2}{x})$.

   (b) $\lim_{x\to 0} \dfrac{\sin 3x}{\tan 2x}$.

   (c) $\lim_{x\to 0} \dfrac{x\sin x}{1-\cos x}$.